

TYBCA – (SEM – V)

501-Linux Operating System (LOS)

Journal Programs

1. Write a shell script to find greatest amongst three numbers.
2. Write a shell script to find all prime numbers in given range.
3. Write a shell script to find sum of digits of a number.
4. Write a shell script to print fibonacci series upto entered value N.
5. Write a menu driven shell script which accepts basic amount as an input and displays following options.
 - a. Dearness allowance (90% of basic)
 - b. Provident Fund F (12% of basic)
 - c. House Rent Allowance (20% of basic + DA)
 - d. Income tax deducted (5% of basic + DA + HRA)
 - e. Take home salary (basic + DA + HRA – IT)
6. Write a shell script to find file permissions of user, group and others.
7. Write a shell script that accepts two files are identical or not.
8. Write a shell script to display all the words, having length <4 characters, of a file f1.txt
9. Write a shell script to find total number of files and total number of directories in current working directory.
10. Write a shell script to find total number of characters, words and lines of a file. (Do not use *wc* command).
11. Write a shell script which accepts a username and check the entered user is currently logged in or not.
12. Write a shell script to find total number of occurrences of SDJIC in given file. (Please provide necessary validations)
13. Write a shell script which accepts filename as input and reverse individual words from it. (Please provide necessary validations)
14. Write *grep* command to perform following actions:
 - a. Count number of blank lines in file f1.txt
 - b. print all lines containing *sdjic*
 - c. print the lines that starts with *sdjic*.
 - d. Search the files in CPROGRAMS directory which has the string "include"
 - e. print lines having exactly 50 characters in file f1.txt
 - f. Count number of blank lines in file f1.txt
 - g. Display lines having atleast one characters in file f1.txt
 - h. Display lines having *sdjic* text in any case in file f1.txt
 - i. Display line of file f1.txt having exactly 3 characters
 - j. Display lines of file f1.txt which begin with any alphabet
 - k. Display lines whose last word is "UNIX" in file f1.txt
 - l. Display filenames having last character as digit [0-9]
 - m. Display list of filenames that only consist digits
 - n. Display line of file f1.txt which only consist digits
 - o. Display lines of file f1.txt which only consist capital alphabets
 - p. Search all lines in file f1.txt which ends with "."
15. Write *sed* command to perform following tasks
 - a. To print only last line of f1.txt
 - b. To print line number 1-3, 6-7 and 10 of f1.txt
 - c. To print lines beginning with SDJIC of f1.txt

- d. Print three lines starting from fourth line of f1.txt
- e. Print all blank lines of file f1.txt
- f. Print lines having either of “sdjic” or “sdjyc”
- g. Lines beginning with either alphabet or digit
- h. To insert a line “additional line” before every line
- i. To replace every occurrence of | with : of first three lines
- j. To replace every occurrence of “|” with “:” of every line
- k. To remove all the lines having word “fail” from file f1.txt (delete command)

Journal Submission Guidelines:

- 1. Lab journals must be submitted **only in journal files**.
- 2. An **index with proper page numbers** is mandatory.
- 3. The **journal submission date** will be announced during lectures.
- 4. **Two-sided printing is not allowed**.
- 5. Each page header must include **student name, roll number, and subject name**.
- 6. Journals will **not be certified** if the above criteria are not followed.

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